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CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS **DATE: 10/07/2026**

EXPERIMENT 2 -- DDL Commands with Constraints – PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK

AIM:

To create and manage constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE and CHECK using DDL commands in a Flight Reservation System.

PROCEDURE:

1. Open MySQL Workbench and select the Flight Reservation database.
2. Create the required tables for Passenger, Flight and Booking.
3. Add a PRIMARY KEY constraint to uniquely identify records.
4. Add CHECK constraints to restrict the values entered in selected columns.
5. Add a UNIQUE constraint to prevent duplicate values.
6. Create FOREIGN KEY constraints to establish relationships between tables.
7. Execute DESC or SHOW CREATE TABLE commands to verify the constraints.
8. Test the constraints by inserting suitable records and observe the output.

PROGRAM:

1. PRIMARY KEY CONSTRAINT

```
CREATE TABLE Passenger (
Passenger_ID INT,
Passenger_Name VARCHAR(50),
Age INT,
Gender CHAR(1),
Email VARCHAR(100)
);


ALTER TABLE Passenger
ADD PRIMARY KEY (Passenger_ID);


DESC Passenger;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Passenger_ID	int	NO	PRI	NULL	
Passenger_Name	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Gender	char(1)	YES		NULL	
Email	varchar(100)	YES		NULL	

2. CHECK CONSTRAINT – GENDER

ALTER TABLE Passenger

ADD CONSTRAINT chk_gender

CHECK (Gender IN ('M', 'F'));

SHOW CREATE TABLE Passenger;

OUTPUT:

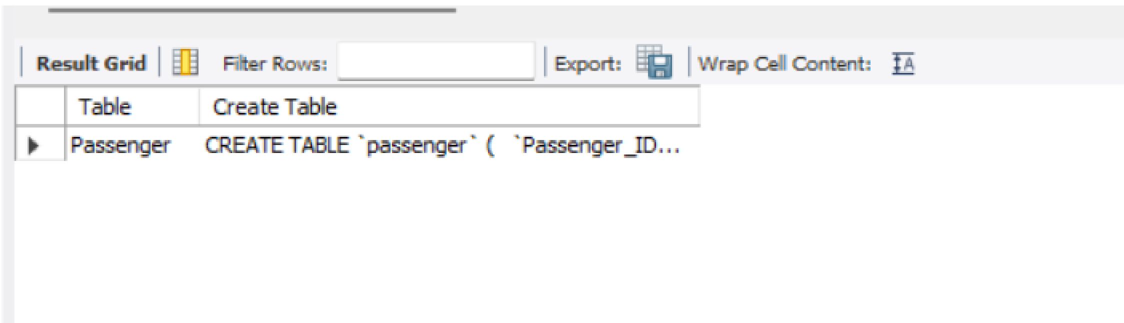


Table	Create Table
Passenger	CREATE TABLE `passenger` (`Passenger_ID...

3. CHECK CONSTRAINT – AGE

ALTER TABLE Passenger

ADD CONSTRAINT chk_age

CHECK (Age > 0);

SHOW CREATE TABLE Passenger;

OUTPUT:

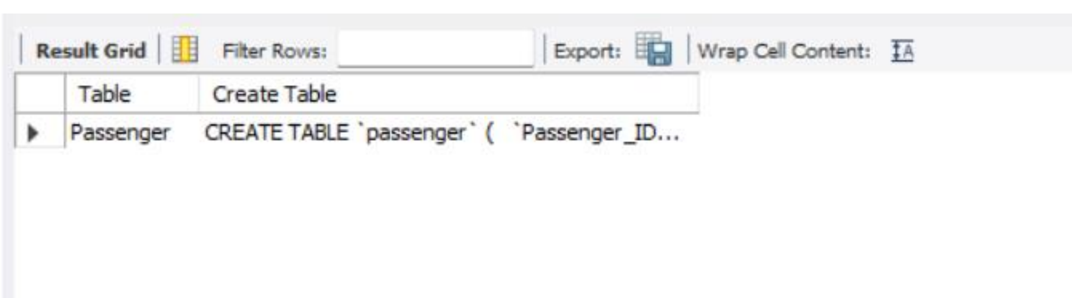
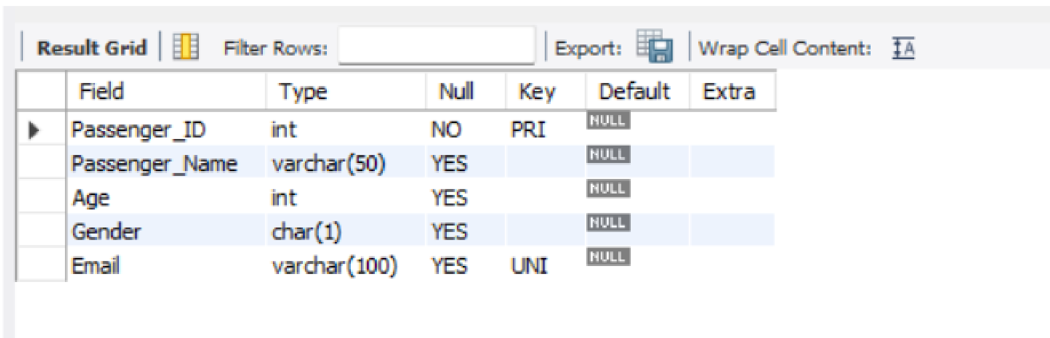


Table	Create Table
Passenger	CREATE TABLE `passenger` (`Passenger_ID...

4. UNIQUE CONSTRAINT

```
ALTER TABLE Passenger
ADD CONSTRAINT unique_email
UNIQUE (Email);
DESC Passenger;
```

OUTPUT:



Field	Type	Null	Key	Default	Extra
Passenger_ID	int	NO	PRI	NULL	
Passenger_Name	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Gender	char(1)	YES		NULL	
Email	varchar(100)	YES	UNI	NULL	

5. CREATE FLIGHT TABLE WITH PRIMARY KEY

```
CREATE TABLE Flight (
   Flight_ID INT PRIMARY KEY,
   Flight_Name VARCHAR(50),
   Source VARCHAR(50),
   Destination VARCHAR(50),
   Price DECIMAL(10,2)
);
DESC Flight;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Flight_ID	int	NO	PRI	NULL	
Flight_Name	varchar(100)	YES		NULL	
Source	varchar(50)	YES		NULL	
Destination	varchar(50)	YES		NULL	
Departure_Time	time	YES		NULL	
Price	decimal(10,2)	YES		NULL	

6. FOREIGN KEY CONSTRAINT

```
CREATE TABLE Booking (
   Booking_ID INT PRIMARY KEY,
   Passenger_ID INT,
   Flight_ID INT,
   Booking_Date DATE
);


ALTER TABLE Booking
ADD CONSTRAINT fk_passenger
FOREIGN KEY (Passenger_ID)
REFERENCES Passenger(Passenger_ID);


ALTER TABLE Booking
ADD CONSTRAINT fk_flight
FOREIGN KEY (Flight_ID)
REFERENCES Flight(Flight_ID);
SHOW CREATE TABLE Booking;
```

OUTPUT:

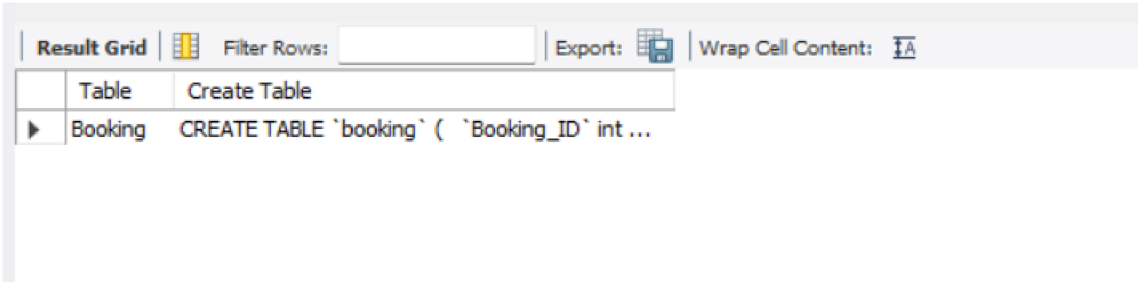


Table	Create Table
Booking	CREATE TABLE `booking` (`Booking_ID` int ...

RESULT:

Thus the DDL commands with PRIMARY KEY, FOREIGN KEY, UNIQUE and CHECK constraints were successfully executed and verified in the Flight Reservation System.